

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the working principle of EDM. Give its advantages and limitations also.

Q.24 Enumerate ten applications of CNC machines.

Q.25 Explain any five robotic sensors.

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Roll No.

4th Sem / Automobile

Subject : CNC Machines and Advanced Manufacturing Processes

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 CNC stands for

- a) Computer Network Control
- b) Computer Numerical Control
- c) Computer Nested Control
- d) Computer Numerical Controller

Q.2 What is the function of the GUI (Graphical User Interface)

- a) To control motion and speed
- b) Converts program into the action of the driver
- c) To record the data from the sensor
- d) To understand the position of the tool according to the input program

- Q.3 Computerizes numerical control machines use
- a) Digital data & a computer
 - b) Computer and a CAM program
 - c) A computer, a part program & a digital data
 - d) None of the above

- Q.4 In USM, material removal occurs due to
- a) abrasion action b) Corroding action
 - c) Chemical action d) Chip formation

- Q.5 The cutting tool in a milling machine is mounted on
- a) Spindle b) Arbor
 - c) Column d) Knee

- Q.6 Which is not the type of part programming format?
- a) Fixed block format
 - b) Variable block format
 - c) Tab sequential format
 - d) Word address format

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 _____ code is used for cutter radius compensation right.

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- Q.8 G04 code is used to specify_____.

- Q.9 Write the full form of ECM.

- Q.10 Open loop control is_____accurate as compared to close loop control system.

- Q.11 Write the purpose of tool offset.

- Q.12 The milling cutters are revolving tools having _____cutting edges.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between absolute coordinate system and incremental coordinate system.

- Q.14 Discuss various types coordinate system in CNC.

- Q.15 Describe any five M code in part programming.

- Q.16 Explain close loop control system in CNC machines.

- Q.17 Write any five advantages of VSM.

- Q.18 Enlist the five applications of ECM.

- Q.19 Give the comparison B/w NC and CNC machines.

- Q.20 Name any four industrial applications of Robots.

- Q.21 Explain the law of robotics.

- Q.22 Describe the working of CNC milling machine.

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